BK7252N Data Sheet

DS-BK7252N-C02 V1.1

2024/7/25

IP Camera

ÿ DTIM10 250 ÿA ÿ

Integrated 200 mA charging management

Endoscope

ÿ 720p resolution

ÿ 544 KB RAM for smooth and clear viewing

Drones

ÿ SPI and DVP dual camera

ÿ 720p resolution

Driving Recorder

ÿ 720p supports 30 FPS

ÿ 120 MHz QSPI display

Table of contents

1. Features

Wi-Fi

- Compliant with IEEE 802.11b/g/n 1x1
- Supports 20 MHz channels
- Support STBC
- Support STA and SoftAP modes
- Support SoftAP + STA coexistence
- Transmit power +17.5 dBm
- Receive sensitivity -98 dBm

Bluetooth Low Energy

- Bluetooth Low Energy 5.2 (Bluetooth LE)
- Supports Bluetooth low energy 1 Mbps, 2 Mbps and long range (125 kbps and 500 kbps)
- Supported Bluetooth low energy features: 2 Mbps, Advertising Extensions, and Long Range

Kernel

- 32-bit MCU, up to 160 MHz
- UART flash download
- SPI flash download
- JTAG debug interface

Memory

- SiP flash capacity up to 4 MB
- On-chip 544 KB RAM
- 4 Bytes eFuse

Clock Management

- External oscillator: 26 MHz crystal oscillator (XTALH), 32 kHz crystal oscillator (XTALL) (QFN40)
- Internal oscillators: 26–160 MHz digitally controlled oscillator (DCO), 32 kHz ring oscillator (ROSC)
- 480 MHz PLL (DPLL)

Power Management

- 2.7 to 4.35 V VCCBAT power supply
- 4.3 to 5.5 V VCCUSB power supply
- On-chip Power-On Reset (POR) and Brown-Out Detect (BOD)
- Built-in LDO regulator
- 200 mA Battery Charger
- Low power consumption:
  - Active mode RX: 42 mA
  - Active mode TX @ 17 dBm: TBD mA
  - Sleep mode: TBD ÿA
  - Shutdown mode: TBD ÿA

Peripherals

- GPIO: 38 for QFN68, 29 for QFN48, 27 for QFN40
- 2 SPI: 1 supports master and slave modes, 1 supports slave mode
- 1 QSPI
- 3 UARTs: 1 with hardware flow control and flash download support
- 1 SDIO
- 2 I2C
- 1 General Purpose DMA Controller (GDMA), 8 channels
- 1 JPEG hardware encoder
- 1 8-bit CIS DVP interface
- 6 32-bit PWM channels
- 1 I2S
- 1 4-band digital hardware equalizer
- 1 audio ADC
- 1 Audio DAC
- 10-bit AUX ADC, up to 7 channels
- Six 32-bit general-purpose timers
- 1 watchdog timer
- 1 real-time counter (RTC)
- 1 infrared port
- 1 temperature sensor

- 1 True Random Number Generator (TRNG)

Encapsulation

- QFN68 package, 8 x 8 mm
- QFN48 package, 6 x 6 mm
- QFN40 package, 5 x 5 mm
- Operating temperature range: -40 to +85 °C

application

- Home appliances
  - refrigerator
  - air conditioner
  - Thermostat
  - washing machine
  - Sweeping robot
- Smart plug
- Smart lighting
  - Light bulb
  - Light switches
  - Ceiling lights
  - Floor lamp
- other
  - Remote control
  - Toy
  - Drones
  - Industrial Terminal
  - Factory Automation Sensors/Switches
  - Smart Meter
  - Payment Terminal
  - Industrial Computer
  - Medical equipment
  - Kitchen appliances
  - Home Automation Switches/Sensors

- Door locks

- Door Camera

2. Overview

The BK7252N is a highly integrated single-chip Wi-Fi 802.11b/g/n and Bluetooth Low Energy (LE) 5.2 combo solution designed for applications that require low power consumption and rich

BK7252N integrates a powerful 32-bit MCU and a complete set of peripherals and interfaces, making it a high-end IoT

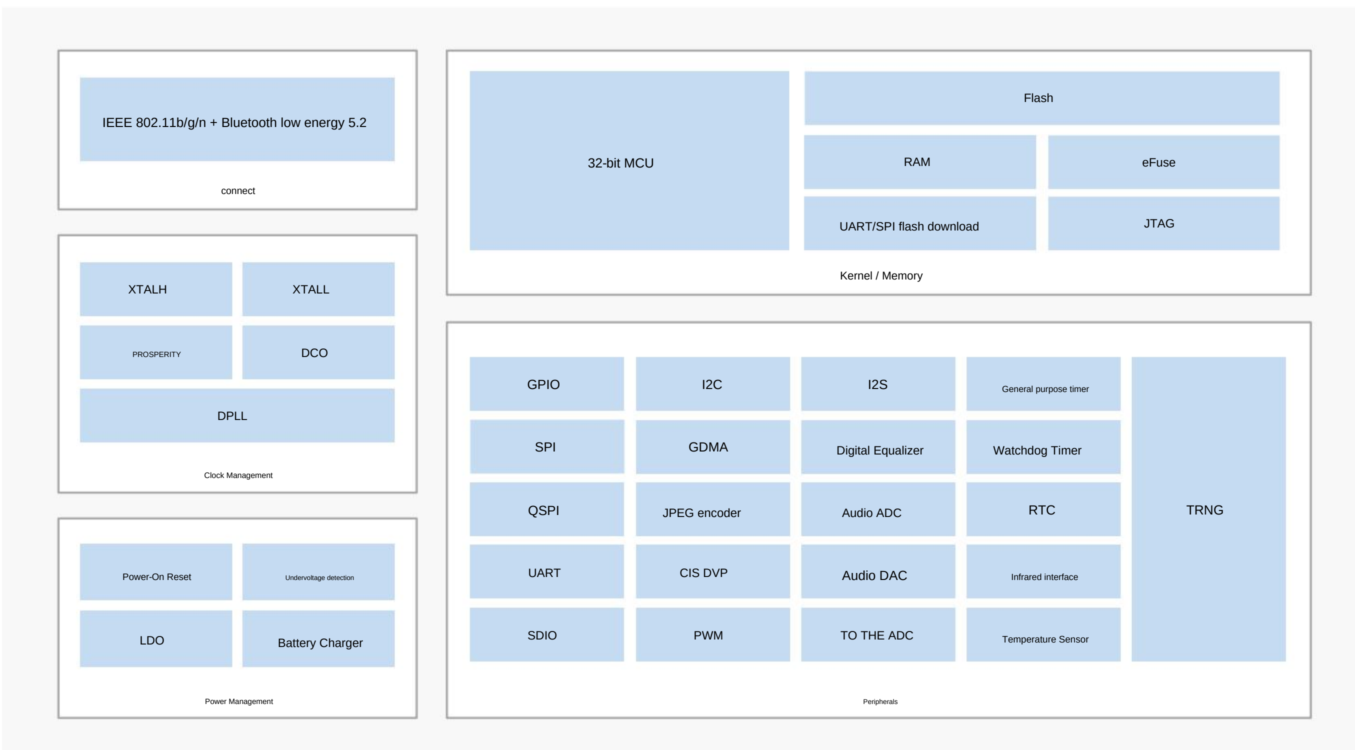
Ideal for IoT (Internet of Things) applications.

BK7252N adopts advanced design technology and process technology to provide a powerful solution for IP cameras, endoscopes, drones, driving recorders and other advanced IoT

and automotive applications with high integration and lowest power consumption.

Figure 2-1 shows the overall block diagram of the BK7252N.

Figure **2-1 BK7252N** block diagram



The BK7252N device is available in three packages. The set of included peripherals varies depending on the package. Table 2-1 shows the list of peripherals available on each package.

Table **2-1** Device options and features

characteristic	QFN68	QFN48	QFN40
SiP flash	2 MB	2 MB	1MB or 2MB
GPIO	38	29	27

characteristic		QFN68	QFN48	QFN40
SPI	Master mode/slave mode	1	1	1
	Slave Mode	1	1	1
QSPI		1	-	-
UART		3	2	2 (Does not support hardware flow control system)
SDIO		1	1	1
I2C	Master mode/slave mode	2	2	2
GDMA		1	1	1
JPEG encoder		1	1	1
CIS DVP Interface		1	1	1
PWM	PWM0–5	4	1	2
I2S	Master mode/slave mode	1	-	-
Digital Equalizer		1	1	1
Audio ADC		1	1	-
Audio DAC		Mono	Mono	-
TO THE ADC	10 digits	1	1	1
	Number of channels	7	6	4
General purpose timer		6	6	6
Watchdog Timer		1	1	1
RTC		1	1	1
Infrared interface		1	-	-
Temperature Sensor		1	1	1
TRNG		1	1	1
Encapsulation		8 x 8 mm QFN68	6 x 6 mm QFN48	5 x 5 mm QFN40
Operating voltage		2.7 to 4.35 V or 4.3 to 5.5 V	2.7 to 4.35 V or 4.3 to 5.5 V	2.7 to 4.35 V
Operating temperature		-40 to +85 °C		

3. Pin Description

The BK7252N offers Wi-Fi and Bluetooth low energy capabilities in three packages ranging from 40-pin to 68-pin.

3.1 QFN68 Pin Description

Figure 3-1 shows the pinout for the 8 x 8 mm, 68-pin QFN package.

Figure **3-1 QFN68** pin assignment

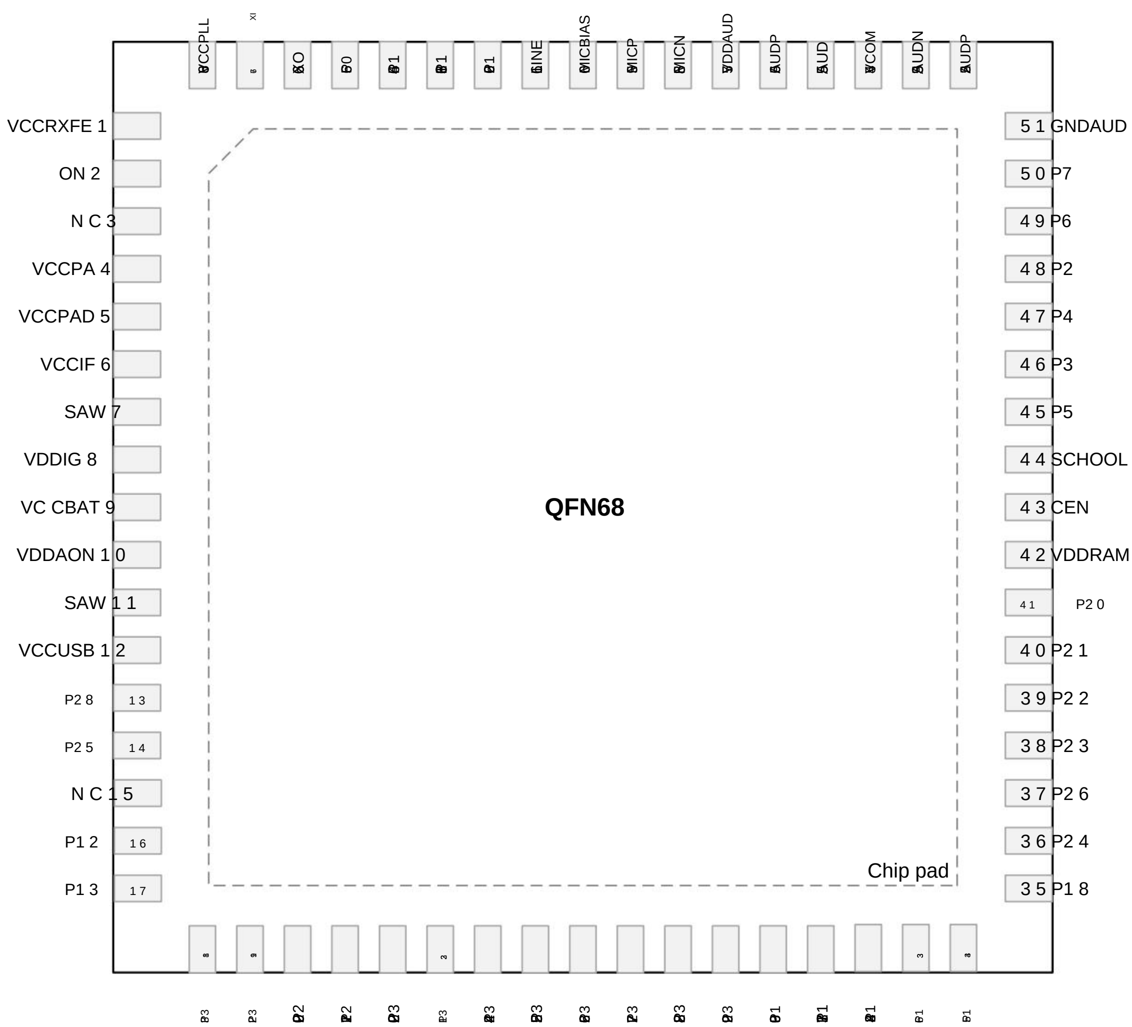


Table 3-1 shows the pin description for the QFN68 package.

Table **3-1 QFN68** pin description

Pin # Name		**I/O** Type		describe
1	VCCRXFE	-	Analog Input	RF Front-End Power Supply
2	ON	-	Radio Frequency	2.4 GHz RF signal port
3	NC	-	NC	No connection
4	VCCPA	-	Analog Input	RF PA Power Supply
5	VCCPAD	-	Analog Input	RF PA Driver Power Supply
6	VCCIF	-	Analog Input	Medium frequency power supply
7	SAW	-	Analog Output	IO LDO Output
8	VDDIG	-	Analog Output	Digital LDO Output
9	VCCBAT	-	power supply	Chip power supply
10	VDDAON	-	Analog Output	AON LDO Output
11	SAW	-	Analog Output	IO LDO Output
12	VCCUSB	-	power supply	Chip power supply
13	P28	I/O Numbers		GPIO28: general purpose I/O ÿ RX_EN: Receive enable
14	P25	I/O Numbers		GPIO25: general purpose I/O ÿ TX_EN: Transmit enable
15	NC	-	NC	No connection
16	P12	I/O	Digital/Analog	GPIO12: general purpose I/O ÿ UART0_RTS: Request to Send ADC6: analog input channel ÿ PGA_INP: Positive input
17	P13	I/O	Digital/Analog	GPIO13: General purpose I/O ÿ UART0_CTS: Clear to send ADC7: Analog input channel ÿ PGA_INN: negative input
18	P33	I/O Numbers		GPIO33: general purpose I/O

Pin # Name		I/O Type		describe
				ÿ CIS_PXD1: Data ÿ SPI0/1_MISO: Master input/slave output
19	P32	I/O Numbers		GPIO32: general purpose I/O ÿ CIS_PXD0: Data ÿ SPI0/1_MOSI: Master output/slave input
20	P29	I/O Numbers		GPIO29: general purpose I/O ÿ CIS_PCLK: pixel clock
21	P27	I/O Numbers		GPIO27: General purpose I/O ÿ CIS_MCLK: master clock
22	P30	I/O Numbers		GPIO30: general purpose I/O ÿ CIS_HSYNC: horizontal synchronization ÿ SPI0/1_SCK: serial clock
23	P31	I/O Numbers		GPIO31: general purpose I/O ÿ CIS_VSYNC: vertical synchronization ÿ SPI0/1_CSN: chip select
24	P34	I/O Numbers		GPIO34: general purpose I/O ÿ CIS_PXD2: Data ÿ SD_CLK: Clock
25	P35	I/O Numbers		GPIO35: general purpose I/O ÿ CIS_PXD3: Data ÿ SD_CMD: Command/Response
26	P36	I/O Numbers		GPIO36: general purpose I/O ÿ CIS_PXD4: Data SD_DATA0: data
27	P37	I/O Numbers		GPIO37: general purpose I/O ÿ CIS_PXD5: Data ÿ SD_DATA1: data
28	P38	I/O Numbers		GPIO38: general purpose I/O ÿ CIS_PXD6: Data

Pin # Name		**I/O** Type		describe
				SD_DATA2: data
29	P39	I/O Numbers		GPIO39: general purpose I/O ÿ CIS_PXD7: Data ÿ SD_DATA3: data
30	P19	I/O Numbers		GPIO19: General purpose I/O ÿ SD_DATA1: data ÿ QSPI_IO3: data
31	P17	I/O Numbers		GPIO17: General purpose I/O SD_DATA0: data ÿ SPI0/1_ MISO: Master Input/Slave Output ÿ QSPI_IO1: data
32	P14	I/O Numbers		GPIO14: general purpose I/O ÿ SD_CLK: Clock ÿ SPI0/1_SCK: serial clock
33	P16	I/O Numbers		GPIO16: general purpose I/O ÿ SD_CMD: Command/Response ÿ SPI0/1_MOSI: Master output/slave input ÿ QSPI_IO0: data
34	P15	I/O Numbers		GPIO15: general purpose I/O SD_DATA3: data ÿ SPI0/1_CSN: chip select
35	P18	I/O Numbers		GPIO18: General purpose I/O SD_DATA2: data ÿ QSPI_IO2: data
36	P24	I/O Numbers		GPIO24: general purpose I/O ÿ LPO_CLK: 32 kHz clock output ÿ PWM4 (differential with PWM5) ÿ QSPI_CLK: serial clock
37	P26	I/O Numbers		GPIO26: general purpose I/O ÿ IRDA: Infrared Data

Pin # Name		**I/O** Type		describe
				ÿ PWM5 (differential from PWM4) ÿ QSPI_CSN: Chip select
38	P23	I/O Numbers		GPIO23: general purpose I/O ÿ DL_SPI_SO: SPI flash download slave output ADC3: Analog input channel ÿ JTAG_TDO: test data output ÿ QSPI_IO0: data
39	P22	I/O Numbers		GPIO22: general purpose I/O ÿ DL_SPI_SI: SPI flash download slave input ÿ CLK26M: 26 MHz clock output ÿ JTAG_TDI: test data input ÿ QSPI_IO1: data
40	P21	I/O Numbers		GPIO21: general purpose I/O ÿ DL_SPI_CSN: SPI flash download chip select ÿ I2C0_SDA: serial data ÿ JTAG_TMS: Test mode selection ÿ QSPI_IO2: data
41	P20	I/O Numbers		GPIO20: general purpose I/O ÿ DL_SPI_SCK: SPI flash download serial clock ÿ I2C0_SCL: serial clock ÿ JTAG_TCK: test clock ÿ QSPI_IO3: data
42	VDDRAM	·	Analog Output	PSRAM Power Supply
43	CEN	·	Analog Input	Chip enable, high level valid
44	GNDIG	·	GND	Grounding
45	P5	I/O	Digital/Analog	GPIO5: general purpose I/O ÿ I2S_DOUT: serial data output ADC2: analog input channel
46	P3	I/O	Digital/Analog	GPIO3: general purpose I/O ÿ I2S_SYNC: frame synchronization

Pin # Name		**I/O** Type		describe
				ADC4: analog input channel ÿ UART2_RX: receive data input
47	P4	I/O	Digital/Analog	GPIO4: General I/O ÿ I2S_DIN: serial data input ADC1: analog input channel
48	P2	I/O	Digital/Analog	GPIO2: General I/O ÿ I2S_CLK: serial clock ADC5: analog input channel ÿ UART2_TX: Send data output
49	P6	I/O Numbers		GPIO6: general purpose I/O ÿ CLK13M: 26 MHz clock output (1/2/4/8 Frequency division ÿ PWM0 (differential from PWM1)
50	P7	I/O Numbers		GPIO7: general purpose I/O ÿ PWM1 (differential from PWM0)
51	GNDAUD	-	GND	Grounding
52	AUDP	-	Analog Output	Audio positive output
53	AUDN	-	Analog Output	Audio negative output
54	VCOM	-	Analog Output	Common mode output
55	AUDN	-	Analog Output	Audio negative output
56	AUDP	-	Analog Output	Audio positive output
57	VDDAUD	-	Analog Output	Audio LDO Output
58	MICN	-	Analog Input	Microphone negative input
59	MICP	-	Analog Input	Microphone positive input
60	MICBIAS	-	Analog Output	Microphone bias output
61	LINE	-	Analog Input	Line in Input
62	P10	I/O Numbers		GPIO10: general purpose I/O

Pin # Name		I/O Type		describe
				ÿ DL_UART_RX: UART flash download receive Data Entry ÿ UART0_RX: receive data input
63	P11	I/O Numbers		GPIO11: general purpose I/O ÿ DL_UART_TX: UART flash download send Data Output ÿ UART0_TX: Send data output
64	P1	I/O Numbers		GPIO1: general purpose I/O ÿ UART1_RX: receive data input ÿ I2C1_SDA: serial data
65	P0	I/O Numbers		GPIO0: general purpose I/O ÿ UART1_TX: Send data output ÿ I2C1_SCL: serial clock
66	XO	·	Analog Output	26 MHz crystal output
67	XI	·	Analog Input	26 MHz crystal input
68	VCCPLL	·	Analog Input	RF PLL Power Supply
Chip pad	GND_SLUG	·	GND	Grounding

3.2 QFN48 Pin Description

Figure 3-2 shows the pinout for the 6 x 6 mm, 48-pin QFN package.

Figure **3-2 QFN48** pin assignment

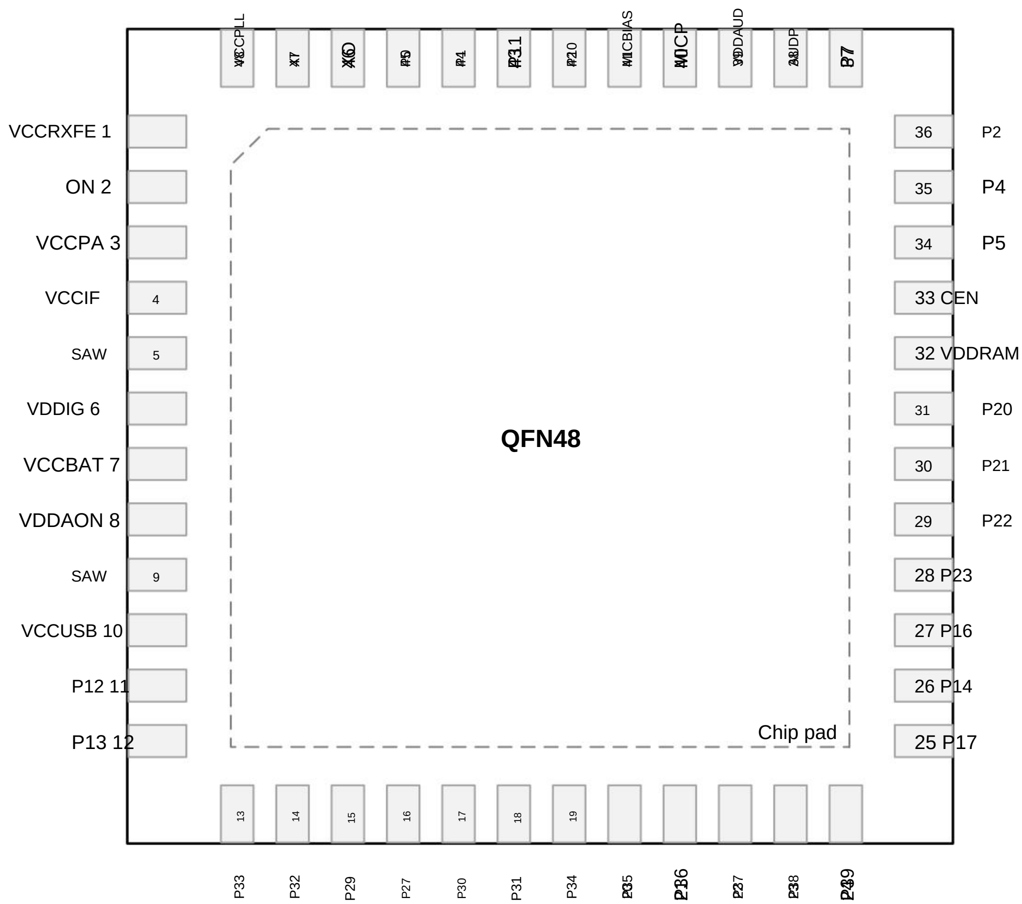


Table 3-2 shows the pin description for the QFN48 package.

Table **3-2 QFN48** pin description

Pin # Name		**I/O** Type		describe
1	VCCRXFE		Analog Input	RF Front-End Power Supply
2	ON		Radio Frequency	2.4 GHz RF signal port
3	VCCPA		Analog Input	RF PA Power Supply
4	VCCIF		Analog Input	Medium frequency power supply
5	SAW		Analog Output	IO LDO Output

Pin # Name		**I/O** Type		describe
6	VDDIG	-	Analog Output	Digital LDO Output
7	VCCBAT	-	power supply	Chip power supply
8	VDDAON	-	Analog Output	AON LDO Output
9	SAW	-	Analog Output	IO LDO Output
10	VCCUSB	-	power supply	Chip power supply
11	P12	I/O	Digital/Analog	GPIO12: general purpose I/O ÿ UART0_RTS: Request to Send ADC6: analog input channel ÿ PGA_INP: Positive input
12	P13	I/O	Digital/Analog	GPIO13: General purpose I/O ÿ UART0_CTS: Clear to send ADC7: Analog input channel ÿ PGA_INN: negative input
13	P33	I/O Numbers		GPIO33: general purpose I/O ÿ CIS_PXD1: Data ÿ SPI0/1_MISO: Master input/slave output
14	P32	I/O Numbers		GPIO32: general purpose I/O ÿ CIS_PXD0: Data ÿ SPI0/1_MOSI: Master output/slave input
15	P29	I/O Numbers		GPIO29: general purpose I/O ÿ CIS_PCLK: pixel clock
16	P27	I/O Numbers		GPIO27: General purpose I/O ÿ CIS_MCLK: master clock
17	P30	I/O Numbers		GPIO30: general purpose I/O ÿ CIS_HSYNC: horizontal synchronization ÿ SPI0/1_SCK: serial clock
18	P31	I/O Numbers		GPIO31: general purpose I/O ÿ CIS_VSYNC: vertical synchronization ÿ SPI0/1_CSN: chip select

Pin # Name		**I/O** Type		describe
19	P34	I/O Numbers		GPIO34: general purpose I/O ÿ CIS_PXD2: Data ÿ SD_CLK: Clock
20	P35	I/O Numbers		GPIO35: general purpose I/O ÿ CIS_PXD3: Data ÿ SD_CMD: Command/Response
21	P36	I/O Numbers		GPIO36: general purpose I/O ÿ CIS_PXD4: Data SD_DATA0: data
22	P37	I/O Numbers		GPIO37: general purpose I/O ÿ CIS_PXD5: Data ÿ SD_DATA1: data
23	P38	I/O Numbers		GPIO38: general purpose I/O ÿ CIS_PXD6: Data SD_DATA2: data
24	P39	I/O Numbers		GPIO39: general purpose I/O ÿ CIS_PXD7: Data SD_DATA3: data
25	P17	I/O Numbers		GPIO17: General purpose I/O SD_DATA0: data ÿ SPI0/1_ MISO: Master Input/Slave Output
26	P14	I/O Numbers		GPIO14: general purpose I/O ÿ SD_CLK: Clock ÿ SPI0/1_SCK: serial clock
27	P16	I/O Numbers		GPIO16: general purpose I/O ÿ SD_CMD: Command/Response ÿ SPI0/1_MOSI: Master output/slave input
28	P23	I/O Numbers		GPIO23: general purpose I/O ÿ DL_SPI_SO: SPI flash download slave output ADC3: Analog input channel

Pin # Name		I/O Type		describe
				ÿ JTAG_TDO: test data output
29	P22	I/O Numbers		GPIO22: general purpose I/O ÿ DL_SPI_SI: SPI flash slave input ÿ CLK26M: 26 MHz clock output ÿ JTAG_TDI: test data input
30	P21	I/O Numbers		GPIO21: general purpose I/O ÿ DL_SPI_CSN: SPI flash download chip select ÿ I2C0_SDA: serial data ÿ JTAG_TMS: Test mode selection
31	P20	I/O Numbers		GPIO20: general purpose I/O ÿ DL_SPI_SCK: SPI flash download serial clock ÿ I2C0_SCL: serial clock ÿ JTAG_TCK: test clock
32	VDDRAM	-	Analog Output	PSRAM Power Supply
33	CEN	-	Analog Input	Chip enable, high level valid
34	P5	I/O	Digital/Analog	GPIO5: general purpose I/O ADC2: analog input channel
35	P4	I/O	Digital/Analog	GPIO4: General I/O ADC1: analog input channel
36	P2	I/O	Digital/Analog	GPIO2: General I/O ADC5: analog input channel
37	P7	I/O Numbers		GPIO7: general purpose I/O ÿ PWM1
38	AUDP	-	Analog Output	Audio positive output
39	VDDAUD	-	Analog Output	Audio LDO Output
40	MICP	-	Analog Input	Microphone positive input
41	MICBIAS	-	Analog Output	Microphone bias output
42	P10	I/O Numbers		GPIO10: general purpose I/O

Pin # Name		I/O Type		describe
				ÿ DL_UART_RX: UART flash download receive Data Entry ÿ UART0_RX: receive data input
43	P11	I/O Numbers		GPIO11: general purpose I/O ÿ DL_UART_TX: UART flash download send Data Output ÿ UART0_TX: Send data output
44	P1	I/O Numbers		GPIO1: general purpose I/O ÿ UART1_RX: receive data input ÿ I2C1_SDA: serial data
45	P0	I/O Numbers		GPIO0: general purpose I/O ÿ UART1_TX: Send data output ÿ I2C1_SCL: serial clock
46	XO	-	Analog Output	26 MHz crystal output
47	XI	-	Analog Input	26 MHz crystal input
48	VCCPLL	-	Analog Input	RF PLL Power Supply
Chip pad GND_SLUG		-	GND	Grounding

3.3 QFN40 Pin Description

Figure 3-3 shows the pinout for the 5 x 5 mm, 40-pin QFN package.

Figure **3-3 QFN40** pin assignment

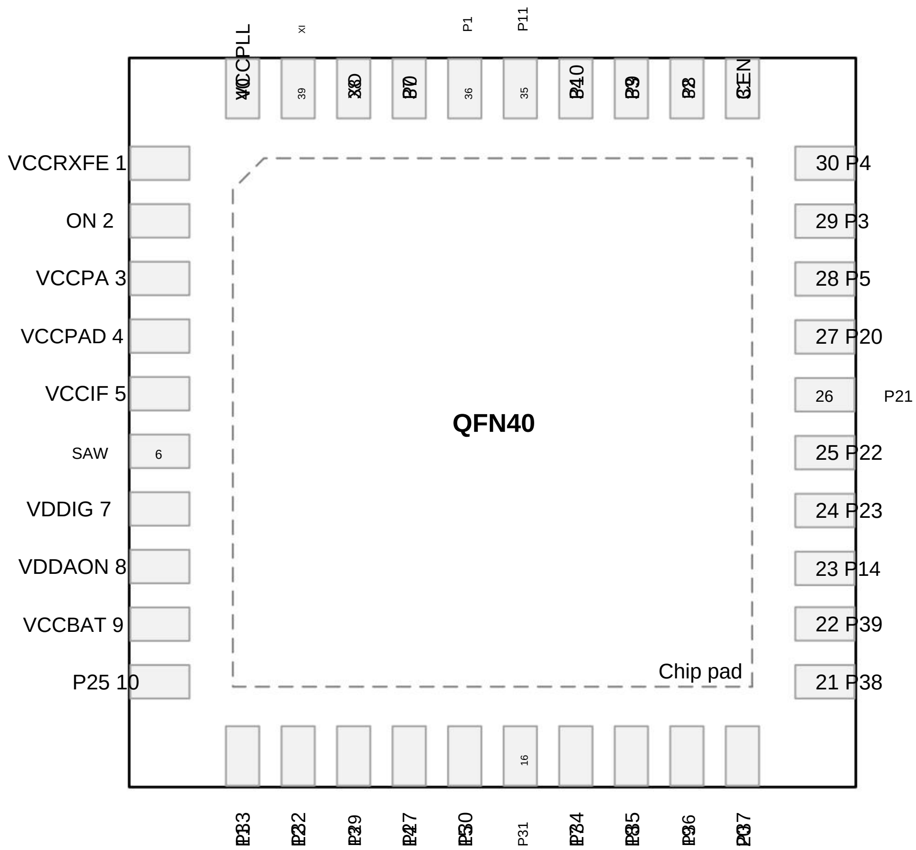


Table 3-3 shows the pin description for the QFN40 package.

Table **3-3 QFN40** pin description

Pin # Name		**I/O** Type		describe
1	VCCRXFE	·	Analog Input	RF Front-End Power Supply
2	ON	·	Radio Frequency	2.4 GHz RF signal port
3	VCCPA	·	Analog Input	RF PA Power Supply
4	VCCPAD	·	Analog Input	RF PA Driver Power Supply
5	VCCIF	·	Analog Input	Medium frequency power supply
6	SAW	·	Analog Output	IO LDO Output
7	VDDIG	·	Analog Output	Digital LDO Output

Pin # Name		I/O Type		describe
8	VDDAON	-	Analog Output	AON LDO Output
9	VCCBAT	-	power supply	Chip power supply
10	P25	I/O Numbers		GPIO25: General purpose I/O
11	P33	I/O Numbers		GPIO33: general purpose I/O ÿ CIS_PXD1: Data ÿ SPI0/1_MISO: Master input/slave output
12	P32	I/O Numbers		GPIO32: general purpose I/O ÿ CIS_PXD0: Data ÿ SPI0/1_MOSI: Master output/slave input
13	P29	I/O Numbers		GPIO29: general purpose I/O ÿ CIS_PCLK: pixel clock
14	P27	I/O Numbers		GPIO27: General purpose I/O ÿ CIS_MCLK: master clock
15	P30	I/O Numbers		GPIO30: general purpose I/O ÿ CIS_HSYNC: horizontal synchronization ÿ SPI0/1_SCK: serial clock
16	P31	I/O Numbers		GPIO31: general purpose I/O ÿ CIS_VSYNC: vertical synchronization ÿ SPI0/1_CSN: chip select
17	P34	I/O Numbers		GPIO34: general purpose I/O ÿ CIS_PXD2: Data ÿ SD_CLK: Clock
18	P35	I/O Numbers		GPIO35: general purpose I/O ÿ CIS_PXD3: Data ÿ SD_CMD: Command/Response
19	P36	I/O Numbers		GPIO36: general purpose I/O ÿ CIS_PXD4: Data SD_DATA0: data
20	P37	I/O Numbers		GPIO37: general purpose I/O

Pin # Name		I/O Type		describe
				ÿ CIS_PXD5: Data ÿ SD_DATA1: data
21	P38	I/O Numbers		GPIO38: general purpose I/O ÿ CIS_PXD6: Data SD_DATA2: data
22	P39	I/O Numbers		GPIO39: general purpose I/O ÿ CIS_PXD7: Data SD_DATA3: data
23	P14	I/O Numbers		GPIO14: general purpose I/O ÿ SD_CLK: Clock ÿ SPI0/1_SCK: serial clock
24	P23	I/O Numbers		GPIO23: general purpose I/O ÿ DL_SPI_SO: SPI flash download slave output ADC3: Analog input channel ÿ JTAG_TDO: test data output
25	P22	I/O Numbers		GPIO22: general purpose I/O ÿ DL_SPI_SI: SPI flash slave input ÿ CLK26M: 26 MHz clock output ÿ JTAG_TDI: test data input
26	P21	I/O Numbers		GPIO21: general purpose I/O ÿ DL_SPI_CSN: SPI flash download chip select ÿ I2C0_SDA: serial data ÿ JTAG_TMS: Test mode selection
27	P20	I/O Numbers		GPIO20: general purpose I/O ÿ DL_SPI_SCK: SPI flash download serial clock ÿ I2C0_SCL: serial clock ÿ JTAG_TCK: test clock
28	P5	I/O	Digital/Analog	GPIO5: general purpose I/O ADC2: analog input channel
29	P3	I/O	Digital/Analog	GPIO3: general purpose I/O

Pin # Name		I/O Type		describe
				ADC4: analog input channel
30	P4	I/O	Digital/Analog	GPIO4: General I/O ADC1: analog input channel
31	CEN	-	Analog Input	Chip enable, high level valid
32	P8	I/O	Digital/Analog	GPIO8: general purpose I/O ÿ PWM2 (differential with PWM3) ÿ 32K_XI: 32.768 kHz crystal input
33	P9	I/O	Digital/Analog	GPIO9: general purpose I/O ÿ PWM3 (differential from PWM2) ÿ 32K_XO: 32.768 kHz crystal output
34	P10	I/O Numbers		GPIO10: general purpose I/O ÿ DL_UART_RX: UART flash download receive Data Entry ÿ UART0_RX: receive data input
35	P11	I/O Numbers		GPIO11: general purpose I/O ÿ DL_UART_TX: UART flash download send Data Output ÿ UART0_TX: Send data output
36	P1	I/O Numbers		GPIO1: general purpose I/O ÿ UART1_RX: receive data input ÿ I2C1_SDA: serial data
37	P0	I/O Numbers		GPIO0: general purpose I/O ÿ UART1_TX: Send data output ÿ I2C1_SCL: serial clock
38	XO	-	Analog Output	26 MHz crystal output
39	XI	-	Analog Input	26 MHz crystal input
40	VCCPLL	-	Analog Input	RF PLL Power Supply
Chip pad GND_SLUG		-	GND	Grounding

4. Functional Description

4.1 Wi-Fi/Bluetooth transceiver

The BK7252N integrates a high-performance Wi-Fi/Bluetooth transceiver. The transceiver integrates two on-chip balancers (baluns). On the receiving side, the on-chip balancer converts the single-ended (unbalanced) RF signal from the antenna into a differential (balanced) signal, and the low noise amplifier (LNA) amplifies the differential signal to achieve better noise and linearity balance. On the transmitting side, the power amplifier (PA) amplifies the differential signal, and the on-chip balancer converts the differential signal into a single-ended signal to feed the antenna. This allows transmission and reception operations to be achieved with only one ANT pin connected to the antenna. The communication range can be extended by configuring GPIO25 and GPIO28 as TX_EN and RX_EN functions to control the external PA and LNA. The transceiver fully integrates the frequency synthesizer without any external components.

4.2 Clock Management

The main clock sources available for BK7252N are as follows:

- High frequency clock
  - 26 MHz crystal oscillator (XTALH)
  - 26–160 MHz internal digitally controlled oscillator (DCO)
  - 480 MHz Digital PLL (DPLL)
- Low frequency clock
  - 32 kHz internal ring oscillator (ROSC)
  - 32 kHz (32.768 kHz) crystal oscillator (XTALL) (QFN40)

The system generates a low power clock source LPO_CLK for standby. LPO_CLK can be selected from the following clocks:

- 32 kHz clock signal derived from a 26 MHz crystal oscillator
- 32 kHz internal oscillator ROSC
- 32 kHz Crystal Oscillator XTALL (QFN40)

BK7252N also has a clock output function, allowing the clock signal to be output to external components through GPIO. GPIO can output the following clock signals:

- CLK26M: high frequency crystal clock, typically 26 MHz
- CLK13M: Clock derived from 26 MHz crystal clock (division factor 1/2/4/8)
- LPO_CLK: LPO_CLK clock
- CIS_MCLK: Reference clock for external CMOS image sensor (CIS)

4.3 Reset

Reset can be triggered by the following sources: power-on reset, brown-out reset, watchdog reset, and wake-up from shutdown mode or deep-sleep mode.

Power-on reset, brown-out reset, and watchdog reset have the same reset effect on all modules, except the normally-on logic. Any of these three resets can reset the entire chip to its initial state. The normally-on logic has a 64-bit timer and a 16-bit retention register, which can only be reset by a power-on reset or Brown-out reset resets to initial value.

Wake-up from shutdown mode triggers a reset of the entire system, while wake-up from deep-sleep mode triggers a reset of the digital blocks.

4.4 Power Management

4.4.1 Power supply solution

The power management system of the BK7252N includes multiple internal LDO regulators to provide voltage and noise isolation for various parts of the chip.

The chip can be powered by VCCBAT from 2.7 to 4.35 V or VCCUSB from 4.3 to 5.5 V. VCCBAT or VCCUSB generates VIO through the IO LDO regulator. VIO is the input power supply for other LDOs.

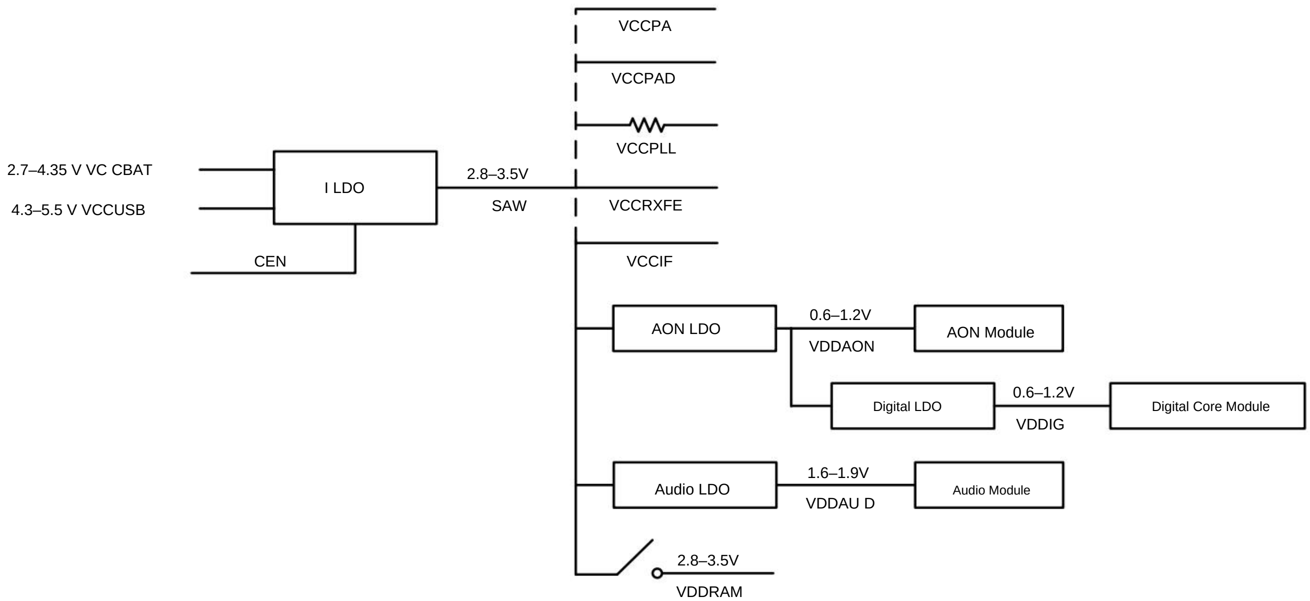
When only VCCBAT is present, the IO LDO regulator is powered by VCCBAT. Similarly, when only VCCUSB is present, the IO LDO regulator is powered by If both VCCBAT and VCCUSB are present, the IO LDO regulator is powered by VCC is powered by USB; when VCCBAT is above 3.0 V, the IO LDO regulator is powered by VCCBAT.

The LDOs generate the following main supplies:

- VIO: Power supply for RF and analog modules and PSRAM
  - Externally connect VCCPA/VCCPAD/VCCPLL/VCCRXFE/VCCIF to power the Wi-Fi/Bluetooth transceiver, and directly power the DPLL, XTAL, AUX ADC, AUDIO power supply
  - Connect to VDDRAM through a switch to power PSRAM
- VDDDIG: Digital logic power supply, powering the processor, memory, Wi-Fi and Bluetooth baseband, and various peripherals
- VDDAON: Always-on (AON) logic power supply. AON logic remains operational in deep sleep mode (VDDDIG powered off), such as RTC and deep sleep to wake-up control logic
- VDDAUD: Audio logic power supply

Figure 4-1 shows the power distribution of BK7252N.

Figure **4-1** Internal power distribution



Note: The output of the LDO regulator requires appropriate bypass capacitors to reduce power supply noise. For more information on selecting bypass capacitors, refer to the Hardware Principle.

Schematic diagram.

Figure 4-2 shows the power-on sequence of BK7252N.

Figure **4-2 BK7252N** power-on sequence

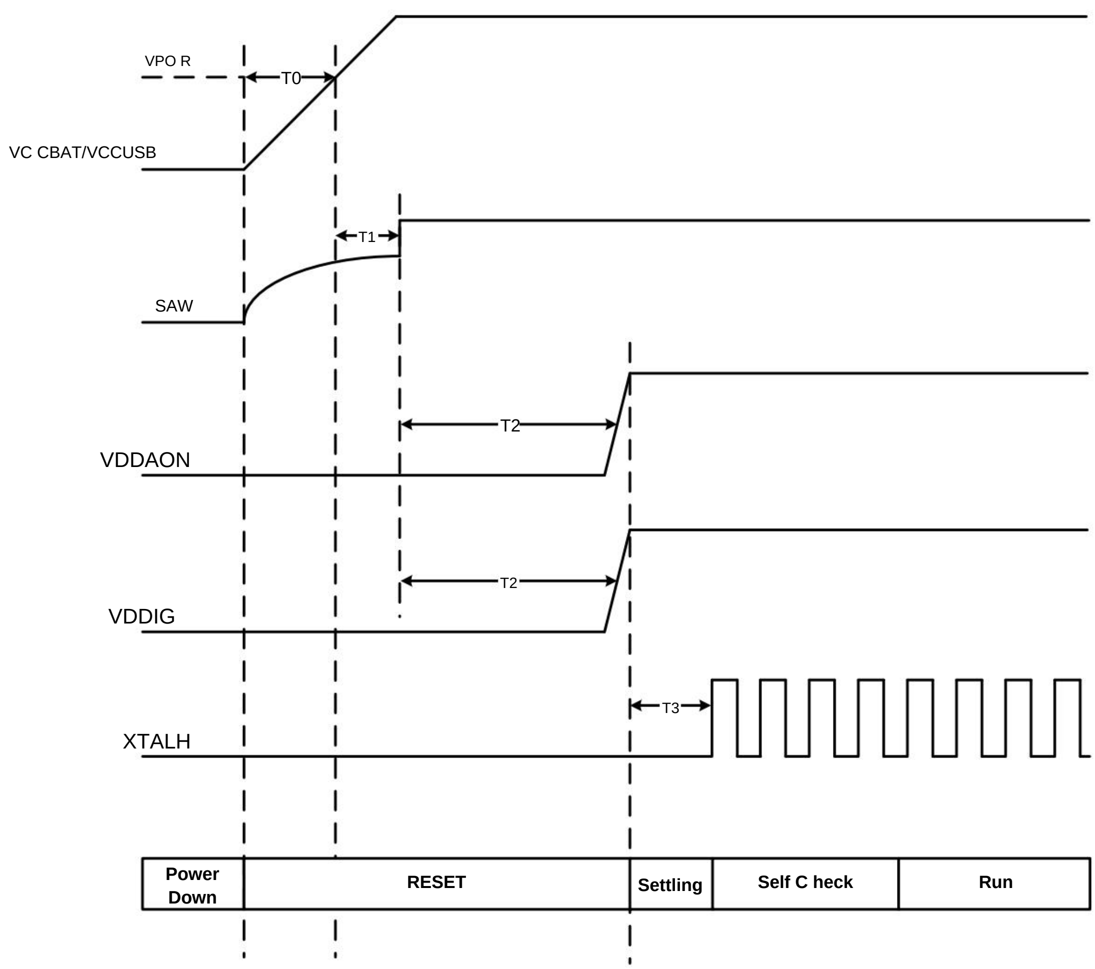


Table **4-1** Timing parameters of power-on sequence

parameter	describe	Min Typ Max Unit			
REBELLION	VCCBAT POR Threshold	·	TBD	·	In
	VCCUSB POR Threshold	·	TBD	·	In
T0	VCCBAT/VCCUSB preparation time	·	TBD	·	ÿs
T1	IO LDO output preparation time	·	TBD	·	ÿs
T2	AON/digital LDO output preparation time	·	TBD	·	ms
T3	XTALH stabilization time	·	TBD	·	ÿs

4.4.2 Battery Charger

When VCCUSB is above 4.3 V, an external battery connected to VCCBAT can be charged by a battery charger.

The charger supports 200 mA constant current charging. The charging current can be programmed in 1.5 mA/step. When the battery reaches the required voltage threshold, the charger switches to constant voltage charging mode.

4.4.3 Power Consumption Mode

BK7252N supports three low power modes besides Active mode, namely shutdown mode, deep sleep mode and sleep mode, among which shutdown mode has the lowest power consumption.

Shutdown mode: All circuits are shut down. A high level on the CEN pin will put the system into Active mode.

Deep Sleep Mode: All circuits are powered off except the normally-on (AON) logic. A GPIO interrupt or RTC interrupt can power the system back on.

The register retains its contents.

Sleep Mode: The MCU and all digital logic stop their clocks and their power supplies are reduced to a much lower holding voltage, resulting in lower current.

The system can be restored to normal voltage by interrupts triggered by the device interrupt, GPIO interrupt, RTC interrupt, or Wi-Fi/Bluetooth MAC low power counter.

Active mode.

Active mode: The MCU is in working state and all peripherals are available.

4.5 General Purpose **I/O (GPIO)**

BK7252N has up to 40 GPIOs, which can be configured as input or output. All GPIOs have alternate functions.

The main features of GPIO include:

- Push-pull output
- Internal pull-up/pull-down resistors
- Configurable drive strength
- Multiplexing function
- Interrupt generation conditions:
    - High or Low
    - Rising or falling edge

4.6 SPI Interface (SPI)

BK7252N integrates two SPI interfaces: SPI0 and SPI1. SPI0 can run in master mode or slave mode and supports clock frequencies up to 60 MHz in both master and slave modes. SPI1 only supports slave mode with clock frequencies up to 60 MHz. SPI0 and SPI1 share SPI pins.

The SPI interface supports the following features:

- 4-wire or 3-wire full-duplex synchronous communication
- Data width:
  - SPI0: configurable 8-bit or 16-bit data width
  - SPI1: 8-bit data width
- Programmable clock polarity and phase (SPI0)
- Programmable data order, supports MSB first or LSB first shifting
- Each SPI interface has a built-in 64-deep RX FIFO and a 64-deep TX FIFO with DMA capability

4.7 QSPI Interface (QSPI)

BK7252N has a built-in Quad SPI interface that supports communication with external flash, PSRAM or AMOLED display. The QSPI interface supports Communication frequency up to 120 MHz.

The characteristics of the QSPI interface are as follows:

- 1-wire, 2-wire or 4-wire SPI input/output
- Two functional modes: indirect mode and memory mapped mode
- Fully programmable opcodes and frame formats
- Integrated RX FIFO and TX FIFO
- Supports 8, 16 and 32-bit data access

4.8 UART Interface (UART)

The BK7252N contains three Universal Asynchronous Receiver/Transmitter (UART) interfaces, supporting full-duplex asynchronous serial communications at baud rates up to 4 Mbps.

The UART interface provides the following features:

- Configurable data length (5, 6, 7 or 8 bits)
- Even, odd or no parity
- Programmable stop bits (1 or 2 bits)

- Each UART has a built-in 128-byte TX FIFO and a 128-byte RX FIFO. FIFO mode is disabled by default and can be enabled by Software enabled.
- Hardware flow control using RTS and CTS signals (UART0) (QFN68 and QFN48)
- Flash Download (UART0)

4.9 SDIO Interface (SDIO)

BK7252N has a secure digital input/output (SDIO) host/slave interface. It can act as a host to read an external SD card and can also be used by an external host.

The maximum clock speed allowed by the SDIO interface is 60 MHz.

SDIO features are as follows:

- Compliant with SD Memory Card Specification Version 2.0
- Compliant with SDIO Card Specification Version 2.0
- Two data bus modes: 1-bit mode (default) and 4-bit mode
- Data transfer up to 30 Mbyte/s in master mode and 20 Mbyte/s in slave mode
- Support DMA function, high-speed transmission without CPU load

4.10 I2C Interface (I2C)

I2C is a popular inter-integrated circuit interface that requires only two buses, the serial data line (SDA) and the serial clock line (SCL).

It forms two I2C interfaces that can operate in master mode or slave mode.

The characteristics of the I2C interface are as follows:

- Master mode and slave mode
- Standard mode (up to 100 kbps)
- Fast mode (up to 400 kbps)
- 7-bit and 10-bit addressing
- Bus Idle and SCL Low Timeout Condition Detection
- Built-in 16-byte TX FIFO and 16-byte RX FIFO

4.11 GDMA Controller (GDMA)

The BK7252N has a general purpose DMA controller (GDMA) with 8 DMA channels to offload CPU activity. These 8 channels are shared by DMA-capable peripherals.

The GDMA controller can perform single block transfers and repeated block transfers. The data width of the destination and source can be configured to 8 bits (byte), 16 bits (half word), or 32 bits (words). The GDMA controller supports high-speed data transfers from peripheral to memory, memory to memory, and memory to peripheral.

A range of peripherals on the BK7252N have DMA capabilities, including UART0, UART1, UART2, SPI0, SDIO, SPI1, JPEG encoding ADC, I2S, AUDIO and AUX ADC.

4.12 JPEG Encoder

The BK7252N contains a JPEG encoder for encoding JPEG streams. The JPEG encoder provides a small hardware compressor for JPEG images. It supports up to 32 programmable quantization tables.

4.13 CMOS Image Sensor Interface (CIS)

The CMOS Image Sensor (CIS) Digital Video Port (DVP) interface provides an 8-bit parallel interface to the sensor, along with the master clock (MCLK), pixel clock (PCLK), horizontal sync (HSYNC), and vertical sync (VSYNC) signals.

The input from the YUV sensor is fed directly into the hardware JPEG encoder, and the output of the JPEG encoder is written directly to the data via a dedicated DMA channel. Memory.

CIS interface features include:

- 8-bit parallel interface
- Programmable polarity of pixel clock and sync signals
- Crop function
- The following data formats are supported:
  - YCbCr 4:2:2 (YUYV, UYVY, YYUV and UVYY)
  - RGB565

4.14 PWM

The BK7252N has three 32-bit PWM pairs, labeled PWM0/1, PWM2/3, and PWM4/5. Each PWM pair consists of two 32-bit incrementing counters. The two counters are driven by an 8-bit programmable prescaler.

Each channel can work independently (arbitrary waveform configuration), or two channels can work in pairs (waveforms are completely opposite and timing is aligned).

The PWM characteristics are as follows:

- 32-bit up counter
- The counter increases in one direction and automatically starts counting again from 0 when it overflows to the maximum value
- Fixed PWM base frequency with 8-bit programmable prescaler (division factor between 1 and 256)

- Six channels, each supporting four modes:
  - PWM mode
  - Timer Mode
  - Counter Mode
  - Capture Mode
- Each channel can be enabled individually and the mode of each channel can be configured individually
- Configurable PWM period and duty cycle for each PWM channel
- In capture mode, it is possible to count continuously between two rising edges, two falling edges, or any two edges
- In timer mode, real-time count value can be read

4.15 I2S Interface (I2S)

BK7252N integrates an I2S interface, supports master mode and slave mode, and supports sampling rates from 8 kHz to 384 kHz. The I2S interface supports PCM Mono mode and I2S stereo channel mode.

I2S features are as follows:

- Master mode or slave mode
- Full-duplex or half-duplex communication
- Support various sampling rates
- 12-bit programmable prescaler
- Support multiple I2S protocols:
  - I2S Philips standard
  - MSB justification (left justification)
  - LSB alignment (right alignment)
  - PCM Standard
- Programmable data order, LSB first or MSB first
- Programmable data width from 1 to 32 bits
- Programmable clock polarity
- Integrated 32-bit RX FIFO and 32-bit TX FIFO, FIFO depth is 32 x 3 channels each

4.16 Audio Peripherals

BK7252N is equipped with rich audio peripherals to enhance the listening experience. The chip includes 4-band digital equalizer, analog-to-digital converter (ADC), digital-to-analog converter (DAC), microphone input amplifier and bias generator, microphone input, line in input, audio output, etc.

4.16.1 4 -band digital equalizer

A dedicated 4-band digital equalizer is implemented before the D/A conversion, allowing the user to customize the frequency response of the audio output. The equalizer is implemented in hardware and can reduce

Overall power consumption of the chip.

4.16.2 Audio ADC and DAC

The BK7252N contains a 16-bit ADC with sampling rates of 8 kHz, 16 kHz, 44.1 kHz or 48 kHz. The chip also integrates a 16-bit DAC with sampling rates of 8 kHz, 16 kHz, 44.1 kHz or 48 kHz.

4.16.3 Microphone Input Amplifier and Bias Generator

The BK7252N has a fully differential analog microphone input amplifier and a low noise microphone bias generator that allows microphones to be connected with passive resistors and capacitors.

4.16.4 Microphone Input

BK7252N provides a microphone input. The microphone signal can be amplified by the microphone input amplifier with a gain range of 0 to 32 dB in steps of 2 dBÿ

4.16.5 Line in Input

BK7252N provides one line in input. The line in input signal can be amplified by the input amplifier with a gain range of 0 to 6 dB in steps of 2 dB. The digitized line in input signal can be further processed using a 4-band equalizer before D/A conversion.

4.16.6 Audio Output

The BK7252N provides a high-fidelity mono audio output capable of driving a 16 ÿ speaker with a load capacitance of up to 30 pF.

4.17 Auxiliary ADC (AUX ADC)

The Auxiliary ADC (AUX ADC) is a 10-bit successive approximation analog-to-digital converter. The AUX ADC has multiple external analog input channels and internal dedicated

The AUX ADC supports analog/digital conversion in single-step mode, software-controlled mode, or continuous mode.

The AUX ADC has the following features:

- Programmable sampling rate from 14.5 kHz to 928.6 kHz
- 10-bit resolution

- Up to seven external analog input channels: ADC1/2/3/4/5/6/7
- Four dedicated internal channels:
  - VBAT monitoring channel (VBAT/3), connected to ADC0
  - TSSIO, connected to ADC8
  - Internal temperature sensor (TEMP), connected to ADC9
  - Internal debug channel, connected to ADC10
- Conversion Mode:
  - Single step mode
  - Software control mode
  - Continuous mode

4.18 Timer Group (TIMG)

BK7252N contains two general purpose timer groups (TIMG). Each group has three 32-bit timers. Each group consists of three 32-bit counters driven by a 4-bit prescaler.

Each TIMG module has the following features:

- Three timers (Timer0/1/2)
- Three 32-bit up-counters
- 4-bit prescaler with division factor between 1 and 16
- Ability to read real-time value of counter

4.19 Watchdog Timer (WDT)

The purpose of the watchdog timer is to detect and recover from faults or malfunctions. It triggers a reset when a specified time period is reached.

The watchdog timer is clocked by the 32 kHz LPO_CLK, and its maximum programmable period is 65.536 (2^16/1 kHz) seconds.

4.20 Real-time Counter (RTC)

The Real-Time Counter (RTC) module has a 64-bit counter and a tick event generator. The RTC is clocked by the 32 kHz LPO_CLK. It is used for low-power timekeeping and can keep running even when the system is in deep sleep mode.

4.21 Infrared interface (IrDA)

BK7252N has built-in hardware IrDA interface, supporting waveform analysis and waveform generation. It monitors the start of infrared signal, records the sequence of infrared waveform, stores the waveform in RX FIFO for software analysis, and writes the waveform to be sent to TX FIFO when sending, thus realizing the analysis and transmission of any infrared protocol.

IrDA has the following features:

- Single and Duplex Mode
- Supports carrier modulation when sending
- Integrated 512-byte RX FIFO and 512-byte TX FIFO

4.22 Temperature Sensor

The BK7252N integrates an on-chip temperature sensor that can measure on-chip temperature in the range of -40 to +125 °C with an accuracy of ±5 °C. The digital result can be to read from the ADC.

Typically, the software will initiate calibration of specific modules based on the temperature value to reduce the difference in chip performance at different temperatures. The host can also read the on-chip temperature and determine Whether to reduce transmit power or suspend operation at high temperatures.

4.23 True Random Number Generator (TRNG)

The random number generator module generates true, non-deterministic random numbers based on thermal noise, which are used to create keys, initialization vectors, and random numbers required for cryptographic operations. number.

5. Electrical characteristics

5.1 Absolute Maximum Ratings

Stresses exceeding the “Absolute Maximum Ratings” may cause permanent damage to the device. Exposure to Absolute Maximum Rating conditions for extended periods may affect the device reliability.

parameter	describe	Minimum Maximum	Unit	
VCCBAT	Chip power supply voltage	-0.3	4.5	In
VCCUSB	USB supply voltage	-0.3	6.0	In
SAW	IO LDO Output Voltage	-0.3	3.8	In
VCCPA	PA supply voltage	-0.3	3.8	In
VCCPAD	PA driver supply voltage	-0.3	3.8	In
VCCIF	IF power supply voltage	-0.3	3.8	In
VCCRXFE	RF RX supply voltage	-0.3	3.8	In
VCCPLL	RF PLL supply voltage	-0.3	3.8	In
VDDRAM	PSRAM supply voltage	-0.3	3.8	In
VDDAON	AON LDO Output Voltage	-0.3	1.3	In
VDDIG	Digital LDO Output Voltage	-0.3	1.3	In
VDDAUD	Audio LDO Output Voltage	-0.3	2.0	In
PRX	Receive input power	.	10	dBm
TSTR	Storage temperature range	-55	150	°C

5.2 ESD Rating

parameter	describe	Test conditions	value	unit
ESD HBM	Electrostatic discharge voltage (Human Body Model HBM), in accordance with ANSI/ESDA/JEDEC JS-001 Standard	.	TBD	In

parameter	describe	Test conditions	value	unit
ESD CDM	Electrostatic discharge voltage (charged device model CDM), compliant ANSI/ESDA/JEDEC JS-002 Standard	-	TBD	In

5.3 Recommended operating conditions

parameter	describe	Min.(1)	Typ.	Max.	Unit
VCCBAT(2)	Chip power supply voltage	2.7	-	4.35	In
VCCBAT Slope	-	TBD	-	-	mV/ms
VCCUSB	USB supply voltage	4.3	5.0	5.5	In
SAW	IO LDO Output Voltage	2.8	-	3.5	In
VCCPA(2)	PA supply voltage	2.8	-	3.5	In
VCCPAD(2)	PA driver supply voltage	2.8	-	3.5	In
VCCIF	IF power supply voltage	2.8	-	3.5	In
VCCRXFE	RF RX supply voltage	2.8	-	3.5	In
VCCPLL	RF PLL supply voltage	2.8	-	3.5	In
VDDRAM	PSRAM supply voltage	2.8	-	3.5	In
VDDAON	AON LDO Output Voltage	0.6	1.15	1.2	In
VDDIG(3)	Digital LDO Output Voltage	0.6	1.15	1.2	In
VDDAUD	Audio LDO Output Voltage	1.6	1.7	1.9	In
MICBIAS	Microphone bias output voltage	1.8	-	2.4	In
TOPR	Operating temperature range	-40	-	85	°C

(1) The specified minimum voltage includes supply voltage ripple and all other transient voltage drops. Care must be taken when operating at minimum voltages.

(2) To ensure WLAN performance, the power supply ripple must be less than Vpp = 100 mV.

(3) In Active mode, the minimum voltage is 1.0 V. In Sleep mode, the minimum voltage is 0.6 V.

5.4 Digital I/O Characteristics

Symbolic parameters		condition	Min Typ Max Unit			
VIH High level input voltage		-	-	TBD	-	In
VIL Low level input voltage		-	-	TBD	-	In
VOH High level output voltage		-	-	TBD	-	In
VOL Low level output voltage		-	-	TBD	-	In
IDRV	I/O output drive strength	-	-	TBD	-	m.a.
RPU weak pull-up resistor		-	-	TBD	-	kÿ
RPD weak pull-down resistor		-	-	TBD	-	kÿ

5.5 Battery Charger

parameter	describe	Min Typ Max Unit			
VCCUSB	Charger input voltage	4.3	5.0	5.5	In
ITrickle	The trickle mode charging current accounts for the fast charge mode charging current percentage	-	10	20	%
IFast	Fast charge mode charging current	1.5	-	200	m.a.
VEnd (after calibration)	After charging is completed, VCCBAT voltage	-	4.2	-	In

5.6 I LDO

parameter	describe	Min Typ Max Unit			
SAW	IO LDO Output Voltage	2.8	3.3	3.5	In
Load current	-	-	-	500	m.a.

5.7 Digital LDO

parameter	describe	Min Typ Max Unit			
VDDIG	Digital LDO Output Voltage	0.6	1.15	1.2	In
Load current	-	-	-	50	m.a.

5.8 AON LDO

parameter	describe	Min Typ Max Unit			
VDDAON	AON LDO Output Voltage	0.6	1.15	1.2	In
Load current	-	-	-	50	m.a.

5.9 Audio LDO

parameter	describe	Min Typ Max Unit			
VDDAUD	Audio LDO Output Voltage	1.6	1.7	1.9	In
Load current	-	-	-	5	m.a.

5.10 26 MHz Crystal Characteristics

Symbolic parameters		condition	Min Typ Max Unit			
F0	Nominal frequency	-	-	26	-	MHz
ÿF/F0	Frequency tolerance	25 °C	-10	-	+10	ppm
TC	Frequency over operating temperature range stability	-40 to 105 °C Crystal	-20	-	+20	ppm
		-30 to 85 °C Crystal	-10	-	+10	ppm
CL	Load Capacitance	-	7	7.3	9	pF
TS	Sensitivity	-40 to 105 °C Crystal	-	32	-	ppm/pF
		-30 to 85 °C Crystal	-	17	-	ppm/pF

5.11 Power Consumption

Unless otherwise noted, the test conditions are T = 25 °C, VCCBAT = 3.3 V.

parameter	condition	Min Typ Max Unit			
Active Mode					
RX Current	11b: 11 Mbps DSSS	-	TBD -		m.a.
	11g: 54 Mbps OFDM	-	TBD -		m.a.
	11n: MCS7, HT20	-	TBD -		m.a.
TX Current	11b: 11 Mbps DSSS @ 17 dBm	-	TBD -		m.a.
	11g: 54 Mbps OFDM @ 15 dBm	-	TBD -		m.a.
	11n: MCS7, HT20 @ 14 dBm	-	TBD -		m.a.
Sleep Mode					
Sleep	-	-	TBD -		ÿA
Deep Sleep	-	-	TBD -		ÿA
Shutdown Mode					
Shutdown	-	-	TBD -		ÿA

5.12 WLAN Radio Frequency Characteristics

5.12.1 WLAN receiver characteristics

Unless otherwise noted, the test conditions are T = 25 °C, VCCBAT = 3.3 V.

parameter	condition	Min Typ Max Unit			
generally					
Operating frequency range	-	2412	-	2484 MHz	
Sensitivity					
Sensitivity – IEEE 802.11b	1 Mbps DSSS	-	-98	-	dBm
	2 Mbps DSSS	-	-94	-	dBm

parameter	condition	Min	Typ	Max	Unit
(1024-byte PSDU, 8% PERÿ	5.5 Mbps DSSS	·	-93	·	dBm
	11 Mbps DSSS	·	-90	·	dBm
Sensitivity – IEEE 802.11g (1000-byte PSDU, 10% PERÿ	6 Mbps OFDM	·	-91	·	dBm
	9 Mbps OFDM	·	-91	·	dBm
	12 Mbps OFDM	·	-91	·	dBm
	18 Mbps OFDM	·	-88	·	dBm
	24 Mbps OFDM	·	-85	·	dBm
	36 Mbps OFDM	·	-82	·	dBm
	48 Mbps OFDM	·	-78	·	dBm
	54 Mbps OFDM	·	-76	·	dBm
Sensitivity – IEEE 802.11n (4096-byte PSDU, 10% PERÿ	HT20, MCS0	·	TBD	·	dBm
	HT20, MCS1	·	TBD	·	dBm
	HT20, MCS2	·	TBD	·	dBm
	HT20, MCS3	·	TBD	·	dBm
	HT20, MCS4	·	TBD	·	dBm
	HT20, MCS5	·	TBD	·	dBm
	HT20, MCS6	·	TBD	·	dBm
	HT20, MCS7	·	TBD	·	dBm
Maximum receiving level					
Maximum receive level @ 2.4 GHz	11b: 1, 2 Mbps (8% PER, 1024 bytes)	·	10	·	dBm
	11bÿ5.5ÿ11 Mbpsÿ8% PERÿ 1024 bytes)	·	10	·	dBm
	11gÿ6–54 Mbpsÿ10% PERÿ 1000 bytes)	·	-3	·	dBm
	11nÿMCS0–7ÿ10% PERÿ4096 byte)	·	TBD	·	dBm

parameter	condition		Min Typ Max Unit			
Adjacent Channel Suppression						
Adjacent Channel (±30 MHz) Rejection –IEEE 802.11b (1024-byte PSDU, 8% PER, with the required information specified in the conditions Signal level)	1 Mbps DSSS -74 dBm		·	TBD	·	dB
	2 Mbps DSSS -74 dBm		·	TBD	·	dB
Adjacent Channel (±25 MHz) Rejection –IEEE 802.11b (1024-byte PSDU, 8% PER, with the required information specified in the conditions Signal level)	5.5 Mbps DSSS -70 dBm		·	TBD	·	dB
	11 Mbps DSSS -70 dBm		·	TBD	·	dB
Adjacent Channel (±25 MHz) Rejection –IEEE 802.11g (1000-byte PSDU, 10% PER, with the required information specified in the conditions Signal level)	6 Mbps OFDM -79 dBm		·	TBD	·	dB
	54 Mbps OFDM -62 dBm		·	TBD	·	dB
Adjacent Channel (±25 MHz) Rejection – IEEE 802.11n (4096-byte PSDU, 10% PER, with the required information specified in the conditions Signal level)	HT20, MCS0 -79 dBm		·	TBD	·	dB
	HT20, MCS7 -61 dBm		·	TBD	·	dB
Spurious emissions						
Spurious emissions	< 1 GHz		·	·	TBD dBm	
	> 1 GHz		·	·	TBD dBm	

5.12.2 WLAN transmitter characteristics

Unless otherwise noted, the test conditions are T = 25 °C, VCCBAT = 3.3 V.

parameter	condition	Min Typ Max Unit			
generally					
Operating frequency range	·	2412	·	2484 MHz	
Transmit power					
	1 Mbps DSSS	·	17.5	·	dBm

parameter	condition	Min	Typ	Max	Unit
Transmit Power – IEEE 802.11b (SEM meets standard requirements)	11 Mbps DSSS	-	17	-	dBm
Transmit Power – IEEE 802.11g (EVM complies with standard requirements)	6 Mbps OFDM	-	15	-	dBm
	54 Mbps OFDM	-	14	-	dBm
Transmit Power – IEEE 802.11n (EVM complies with standard requirements)	HT20, MCS0	-	14	-	dBm
	HT20, MCS7	-	13	-	dBm
Spurious emissions					
Spurious emission (maximum output power)	< 1 GHz	-	-	TBD dBm	
	> 1 GHz	-	-	TBD dBm	

5.13 Bluetooth Low Energy Radio Frequency Characteristics

5.13.1 Bluetooth Low Energy Receiver Characteristics

Unless otherwise noted, the test conditions are T = 25 °C, VCCBAT = 3.3 V.

parameter	condition	Min	Typ	Max	Unit
generally					
Operating frequency range	-	2402	-	2480 MHz	
THE 1 Mbps					
Sensitivity	30.8% PER	-	-96	-	dBm
Maximum input level	30.8% PER	0	-	-	dBm
C/I Co-channel	-	-	8	21	dB
C/I 1 MHz adjacent channel	-	-	2	15	dB
C/I -1 MHz adjacent channel	-	-	-3	15	dB
C/I 2 MHz adjacent channel	-	-	-20	-17	dB
C/I -2 MHz adjacent channel	-	-	-31	-17	dB
C/I 3 MHz adjacent channel	-	-	-25	-18	dB

parameter	condition	Min Typ Max Unit			
C/I -3 MHz adjacent channel	-	-	-31	-27	dB
C/I > 3 MHz adjacent channel	-	-	-31	-27	dB
C/I < -3 MHz adjacent channel	-	-	-31	-27	dB
Out-of-band blocking	30–2000 MHz	-30	-	-	dBm
	2003–2399 MHz	-35	-	-	dBm
	2484–2997 MHz	-35	-	-	dBm
	3000 MHz–12.75 GHz	-30	-	-	dBm
Intermodulation	-	-	-	-40	dBm
THE 2 Mbps					
Sensitivity	30.8% PER	-	-94	-	dBm
Maximum input level	30.8% PER	0	-	-	dBm
C/I Co-channel	-	-	9	21	dB
C/I 2 MHz adjacent channel	-	-	2	15	dB
C/I -2 MHz adjacent channel	-	-	-3	15	dB
C/I 4 MHz adjacent channel	-	-	-20	-17	dB
C/I -4 MHz adjacent channel	-	-	-31	-17	dB
C/I 6 MHz adjacent channel	-	-	-28	-27	dB
C/I -6 MHz adjacent channel	-	-	-32	-27	dB
C/I > 6 MHz adjacent channel	-	-	-32	-27	dB
C/I < -6 MHz adjacent channel	-	-	-32	-27	dB
Out-of-band blocking	30–2000 MHz	-30	-	-	dBm
	2003–2399 MHz	-35	-	-	dBm
	2484–2997 MHz	-35	-	-	dBm
	3000 MHz–12.75 GHz	-30	-	-	dBm
Intermodulation	-	-	-	-50	dBm

5.13.2 Bluetooth Low Energy Transmitter Characteristics

Unless otherwise noted, the test conditions are T = 25 °C, VCCBAT = 3.3 V.

parameter		condition	Min Typ Max Unit			
generally						
Operating frequency range		·	2402	·	2480	MHz
Transmit power		·	TBD	6	TBD	dBm
THE 1 Mbps						
In-band emission	±2 MHz offset	·	·	-40	-20	dBm
	±3 MHz offset	·	·	-46	-30	dBm
	>±3 MHz offset	·	·	-47	-30	dBm
Modulation characteristics	ÿf1avg	·	225	259	275	kHz
	ÿf2max	·	185	250	·	kHz
	ÿf2avg/ÿf1avg	·	0.8	0.92	·	·
Carrier frequency offset and drift	n = 0, 1, 2, 3…k maximum value - |fn |		·	6	150	kHz
	n = 2, 3, 4…k maximum value - |f0 – fn |		·	3	50	kHz
	|f1 – f0 |	·	·	2	23	kHz
	|fn – fn-5 | n = 6, 7, 8… k max-		·	2	20	kHz/50 ÿs
THE 2 Mbps						
In-band emission	±4 MHz offset	·	·	-28	-20	dBm
	±5 MHz offset	·	·	-44	-20	dBm
	>±5 MHz offset	·	·	-48	-30	dBm
Modulation characteristics	ÿf1avg	·	450	518	550	kHz
	ÿf2max	·	370	507	·	kHz
	ÿf2avg/ÿf1avg	·	0.8	0.94	·	·
Carrier frequency offset and drift	n = 0, 1, 2, 3…k maximum value - |fn |		·	10	150	kHz
	n = 2, 3, 4…k maximum value - |f0 – fn |		·	3	50	kHz

parameter		condition	Min Typ Max Unit			
	|f1 – f0 |	-	-	2	23	kHz
	n = 6, 7, 8…k max- | fn – fn-5 |		-	2	20	kHz/50 ÿs

5.14 Audio Features

parameter	condition	Min Typ Max Unit			
DAC differential output	600 Ohm load	-	1	-	Vrms
	16 Ohm Load	-	0.8	-	Vrms
DAC differential output THD	0.7 Vrms @ 600 Ohm load	-	-	-80	dB
	0.65 Vrms @ 16 Ohm load	-	-	-80	dB
DAC Output SNR	1 kHz sine wave	-	104	-	dB
DAC sampling rate	-	8	-	48	kHz
ADC SNR	1 kHz sine wave	-	92	-	dB
ADC sampling rate	-	8	-	48	kHz

5.15 AUX ADC Characteristics

parameter	condition	Minimum Typical Maximum			unit
Convert Clock	-	0.2	-	13	MHz
Conversion time	-	-	14	-	cycle
VREF	internal	0.8	1.8	-	In
	external	-	VIO/2	-	In
Input voltage range	-	0	-	VREF*2	In
Input Impedance	-	100	-	-	Mÿ
Input Capacitance(Cs)	-	-	1	-	pF
DNL	-	-1	-	0.5	LSB
INL	-	-2	-	2	LSB

parameter	condition	Minimum Typical Maximum			unit
ENOB	-	-	9.2	-	Bit
SNDR	-	-	57	-	dB
SFDR	-	-	61.4	-	dB
TSTARTUP	-	-	5	-	ÿs
Power consumption	-	-	200	-	ÿA

6. Packaging information

6.1 QFN68 8 x 8 mm package

Figure **6-1 QFN68 8 x 8 mm** package outline

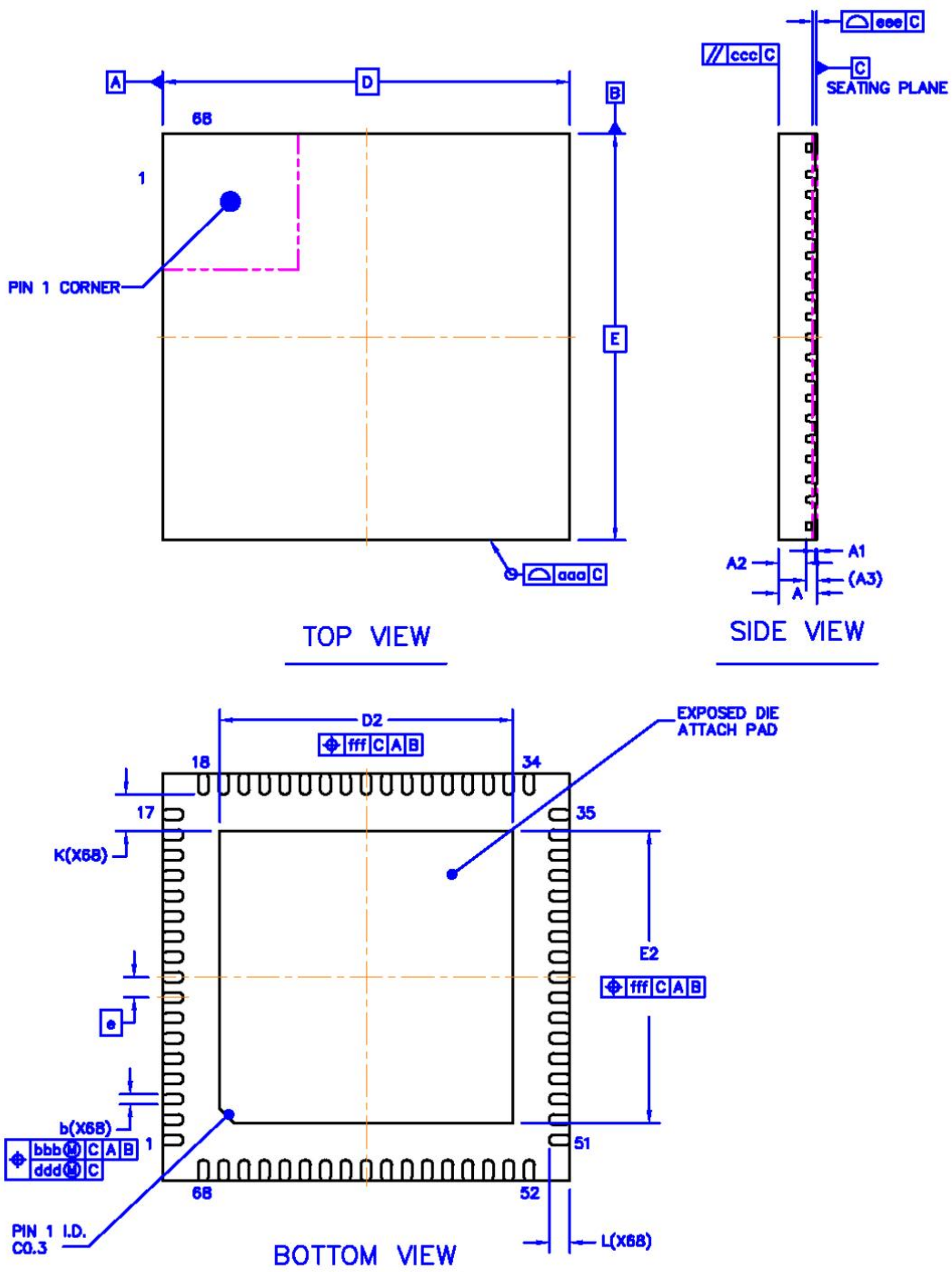


Table **6-1 QFN68** package dimensions

symbol	Dimensions (mm)		
	Minimum	Nominal value	Maximum
A	0.70	0.75	0.80
A1	0.00	0.02	0.05
A2	-	0.55	-
A3	0.203 REF		
b	0.15	0.20	0.25
D	8.00 BSC		
AND	8.00 BSC		
and	0.40 BSC		
D2	5.65	5.75	5.85
E2	5.65	5.75	5.85
L	0.30	0.40	0.50
K	0.725 REF		
aaa	0.10		
ccc	0.10		
eee	0.08		
bbb	0.07		
ddd	0.05		
fff	0.10		

6.2 QFN48 6 x 6 mm package

Figure **6-2 QFN48 6 x 6 mm** package outline

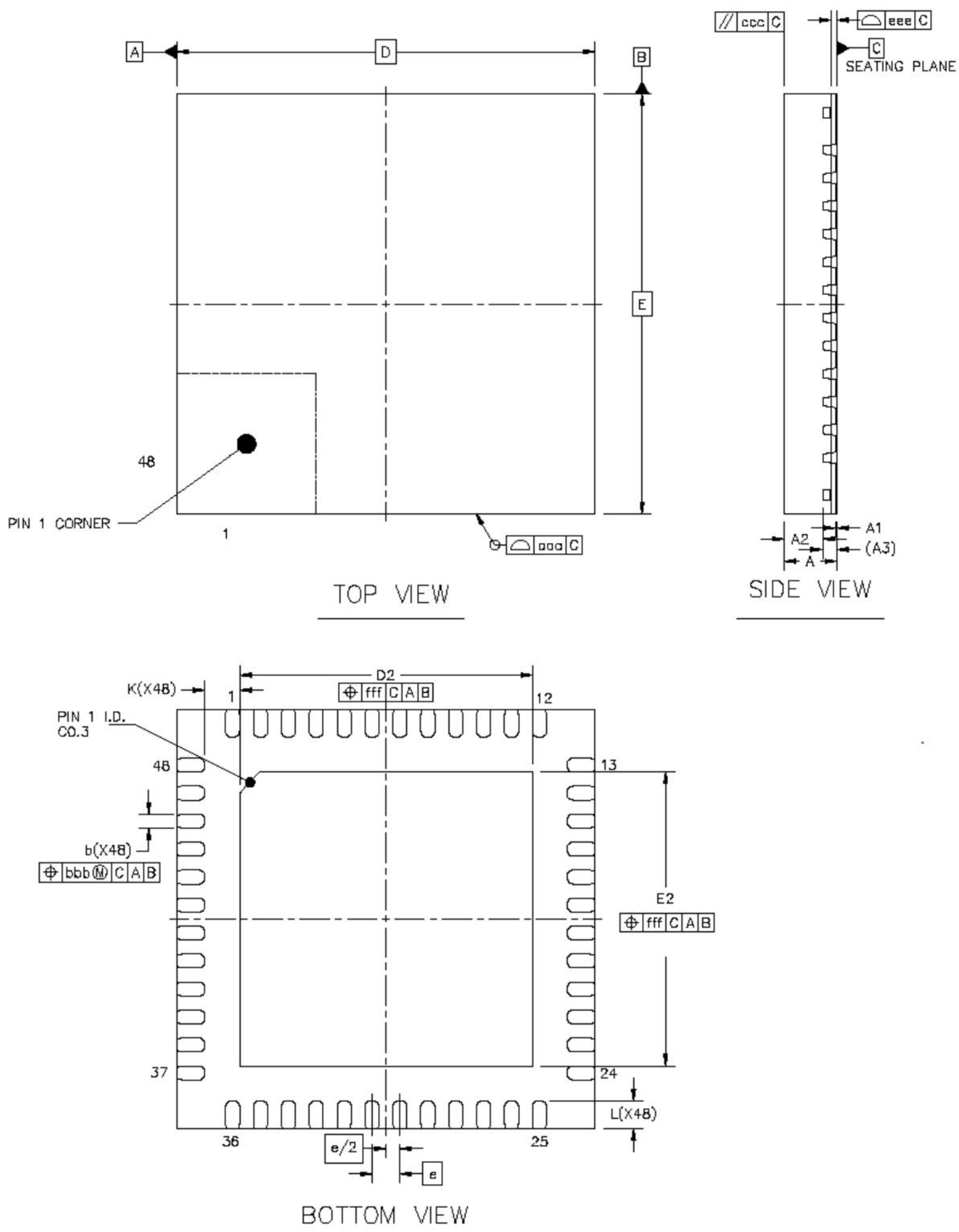


Table **6-2 QFN48** package dimensions

symbol	Dimensions (mm)		
	Minimum	Nominal value	Maximum
A	0.70	0.75	0.80
A1	0.00	0.02	0.05
A2	-	0.55	-
A3	0.203 REF		
b	0.15	0.20	0.25
D	6.00 BSC		
AND	6.00 BSC		
and	0.40 BSC		
D2	4.10	4.20	4.30
E2	4.10	4.20	4.30
L	0.30	0.40	0.50
K	0.50 REF		
aaa	0.10		
ccc	0.10		
eee	0.08		
bbb	0.07		
fff	0.10		

6.3 QFN40 5 x 5 mm package

Figure **6-3 QFN40 5 x 5 mm** package outline

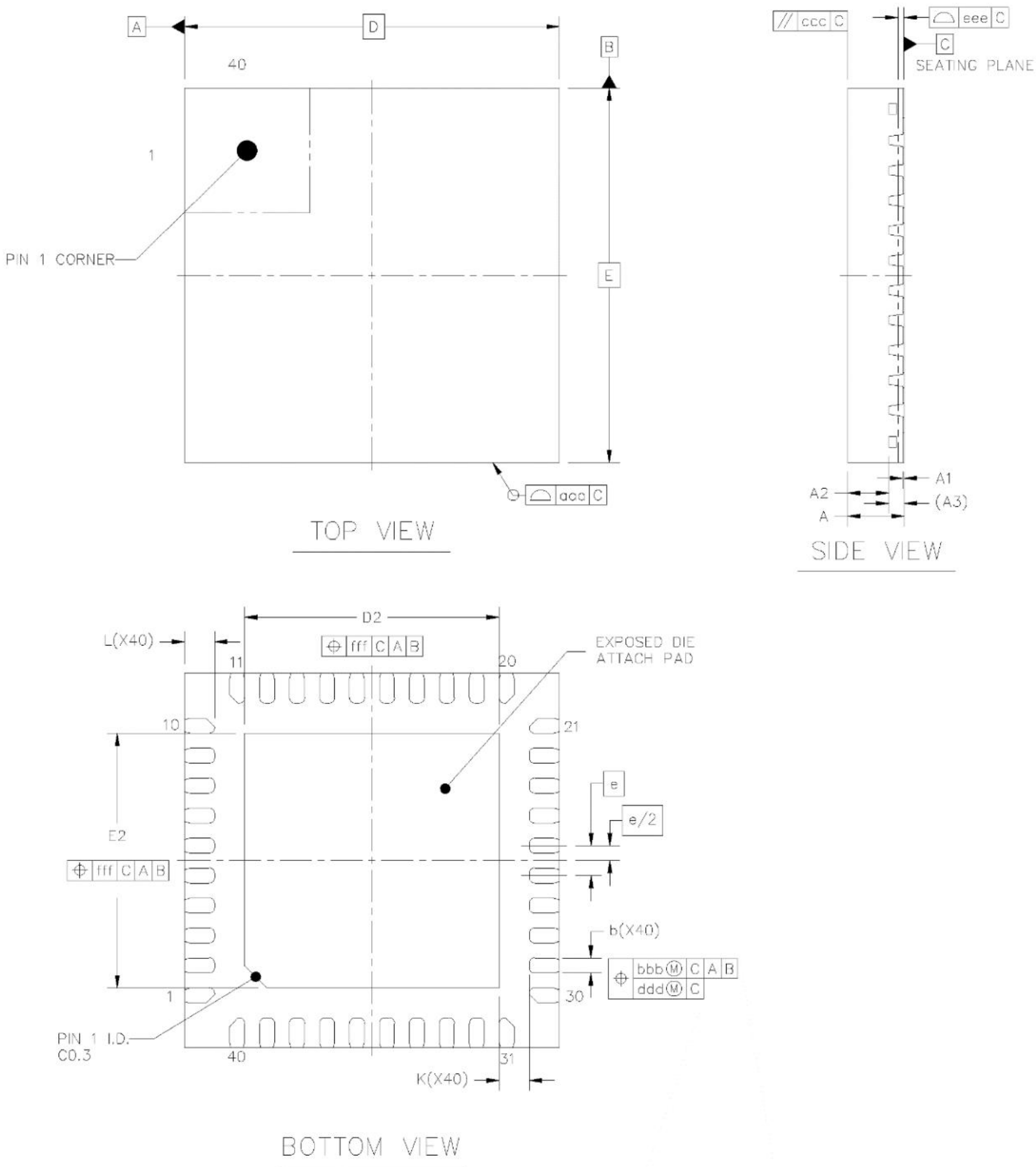
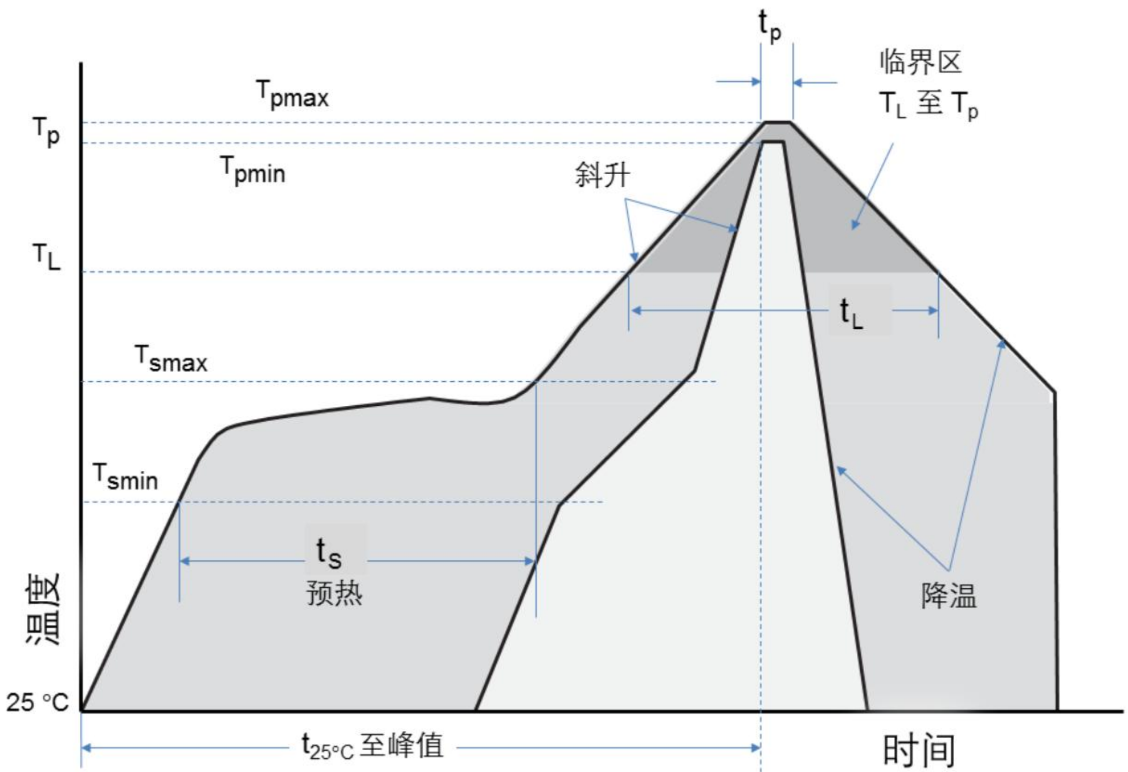


Table **6-3 QFN40** package dimensions

symbol	Dimensions (mm)		
	Minimum	Nominal value	Maximum
A	0.70	0.75	0.80
A1	0.00	0.02	0.05
A2	-	0.55	-
A3	0.203 REF		
b	0.15	0.20	0.25
D	5.00 BSC		
AND	5.00 BSC		
and	0.40 BSC		
D2	3.30	3.40	3.50
E2	3.30	3.40	3.50
L	0.30	0.40	0.50
K	0.40 REF		
aaa	0.10		
ccc	0.10		
eee	0.08		
bbb	0.07		
ddd	0.05		
fff	0.10		

7. Reflow Oven Profile

Figure **7-1** Reflow soldering curve



Curve characteristics		Specification
Average heating rate (Tsmax to Tp)		Maximum 3 °C/sec
Preheat	Minimum temperature (Tsmin)	150 °C
	Maximum temperature (Tsmax)	200 °C
	Time (ts)	60 to 180 seconds
Holding time above this temperature	Temperature (TL)	217 °C
	Time (tL)	60 to 150 seconds
Peak/Classification Temperature (Tp)		260 °C

Curve characteristics	Specification
Time that the actual peak temperature is within 5 °C (tp)	20 to 40 seconds
Cooling rate	Maximum 6 °C/sec
Time between 25 °C and peak temperature	Up to 8 minutes

RoHS compliant

According to the EU RoHS Directive (EU) 2015/863 amending Annex II of Directive 2011/65/EU, this product does not contain lead, mercury, cadmium, hexavalent chromium, PBB, PBDE, DEHP, BBP, DBP and DIBP.

Electrostatic Discharge **(ESD)** Sensitivity

Integrated circuits are ESD sensitive and can be damaged by static electricity. Use proper ESD techniques when handling these devices.

Moisture Sensitivity Level

This product complies with the moisture sensitivity level MSL3 of the IPC/JEDEC J-STD-020 standard.

8. Ordering Information

Figure **8-1** Model naming

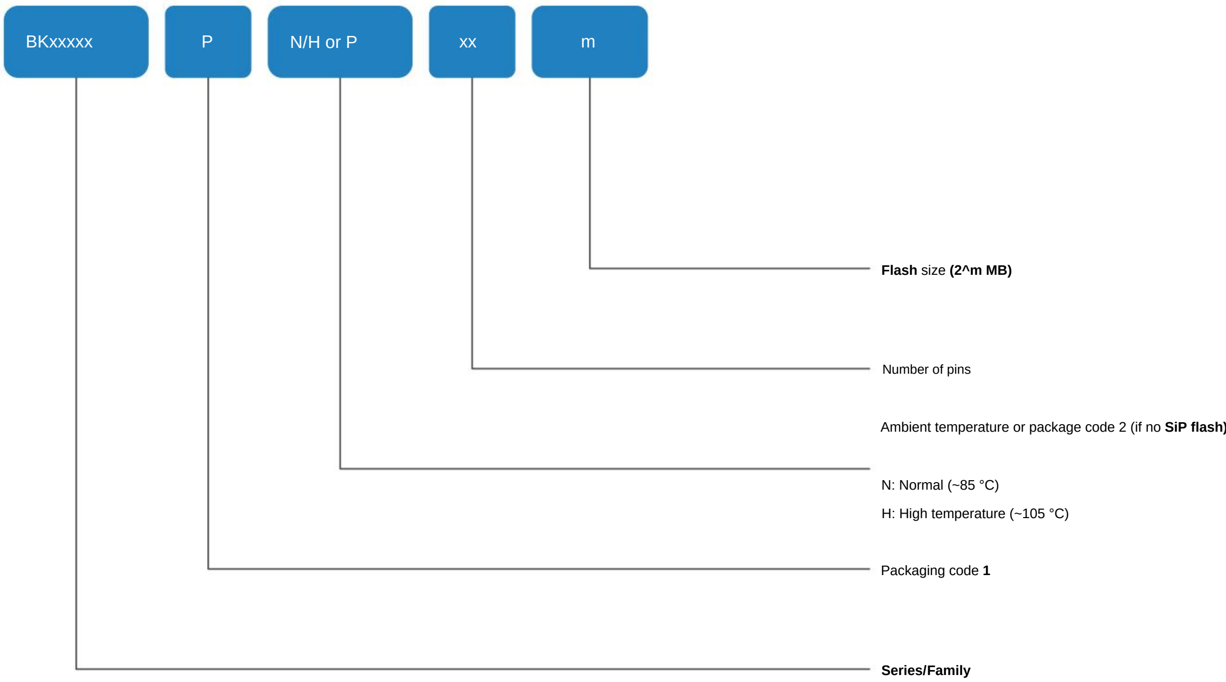


Table **8-1** Ordering information

Order Code	Encapsulation	**SiP(1) flash** ambient temperature		Package	Minimum Order Quantity **(MOQ)**
BK7252NQN681	8 mm x 8 mm QFN68 2 MB		-40 to +85 °C Tape and Reel		3000
BK7252NQN481	6 mm x 6 mm QFN48 2 MB		-40 to +85 °C Tape and Reel		3000
BK7252NQN400 5 mm x 5 mm QFN40 1 MB			-40 to +85 °C Tape and Reel		3000
BK7252NQN401	5 mm x 5 mm QFN40 2 MB		-40 to +85 °C Tape and Reel		3000

(1) SiP flash refers to flash packaged inside a chip.

Revision History

Version	date	Release Notes
1.0	2024/3/5	Initial release
1.1	2024/7/25	ÿ General wording revisions ÿ Updated the maximum operating temperature ÿ QFN40 package with 1 MB SiP flash ÿ Updated the description of Wi-Fi/Bluetooth transceiver in Section 4.1 ÿ Updated the description of reset in Section 4.3 ÿ Added I2C FIFO feature in Section 4.10 I2C Interface (I2C) ÿ Corrected the sampling rate of AUX ADC in Section 4.17 AUX ADC scope ÿ Corrected the conversion clock frequency of AUX ADC in Section 5.15 AUX ADC Characteristics and conversion time ÿ Updated the package information of QFN68 package, QFN48 package and QFN40 package in Section 6 Package Information ÿ Added order code BK7252NQN40 in Section 8 Ordering Information

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